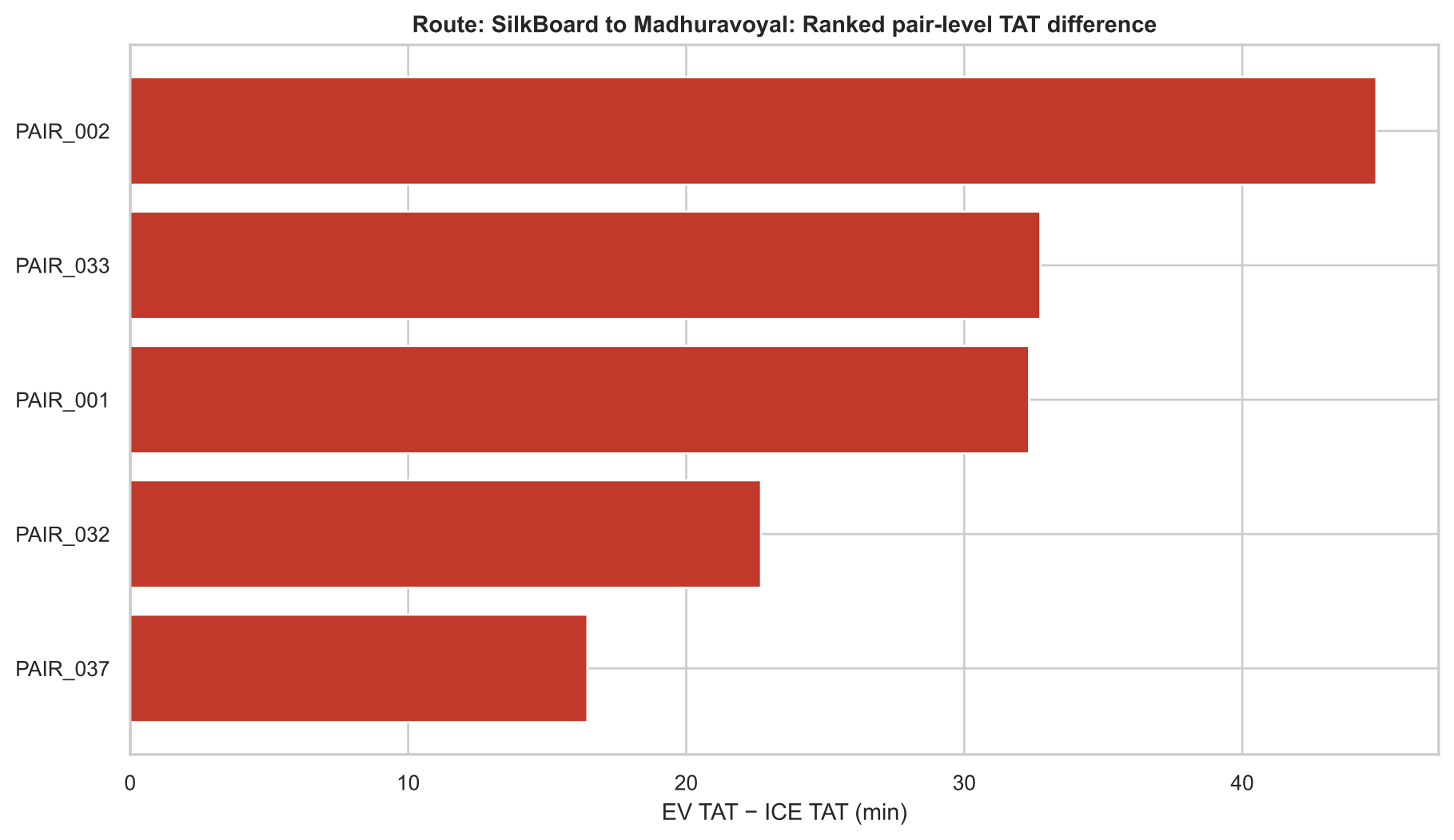
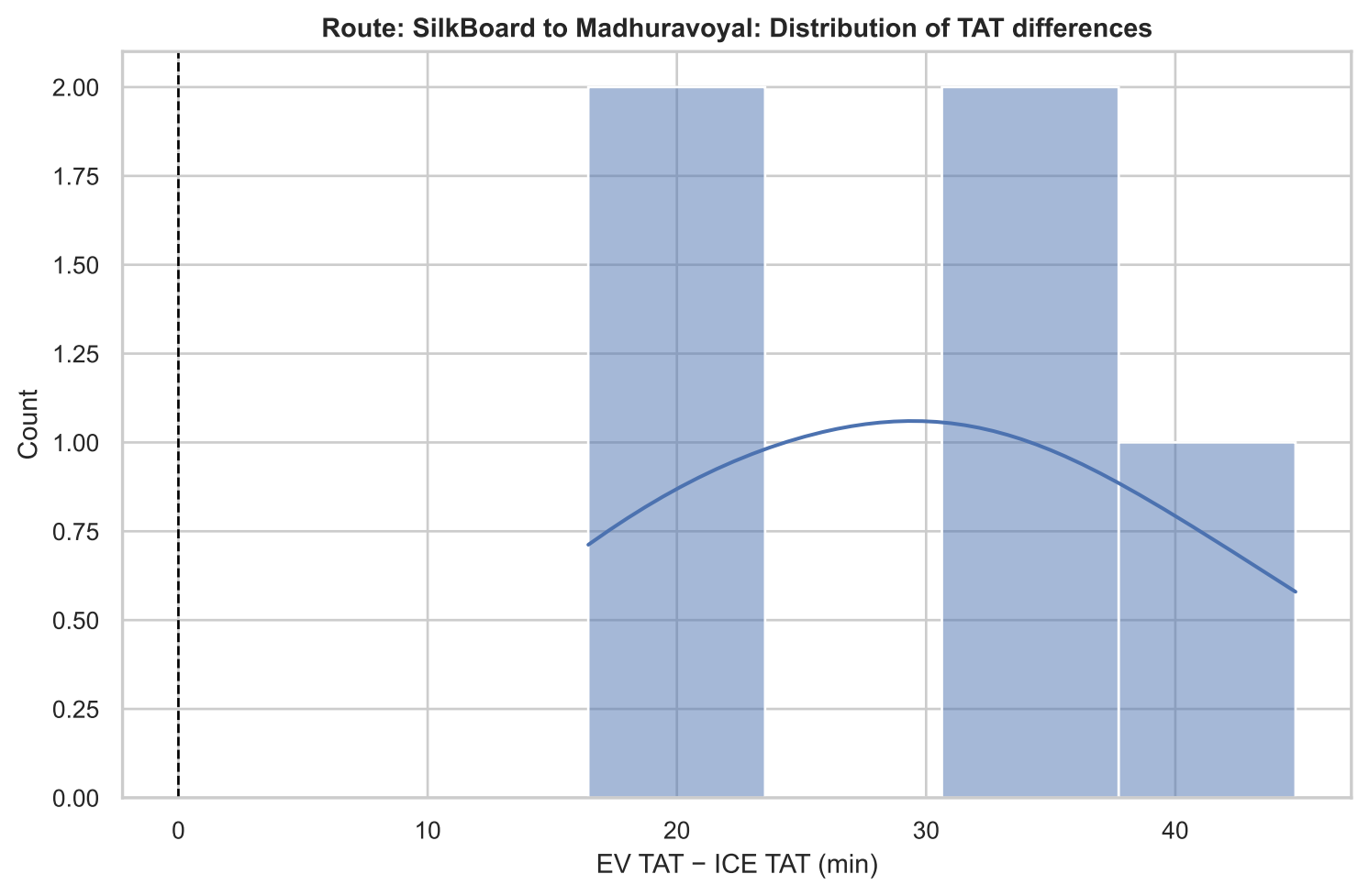
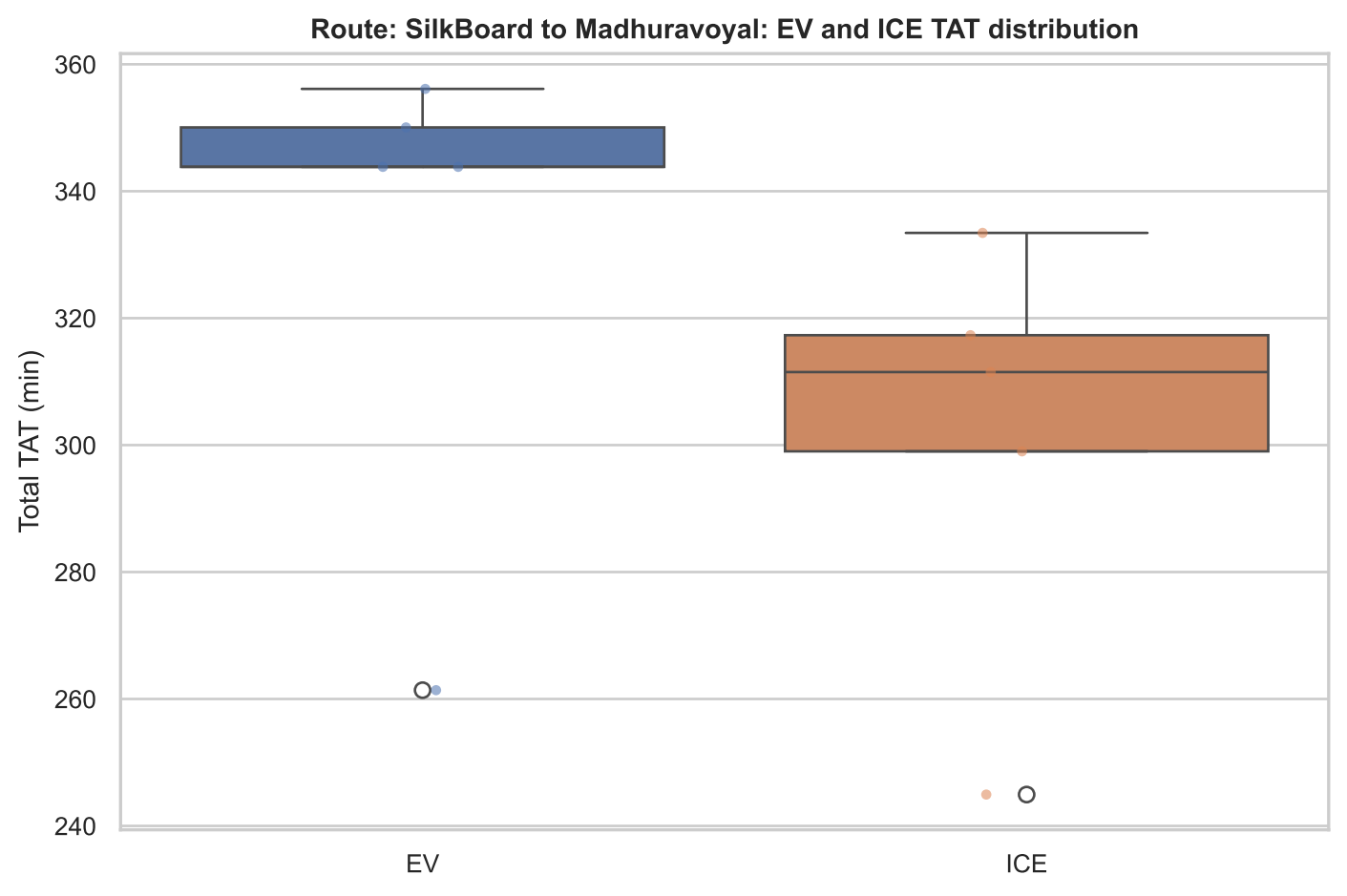
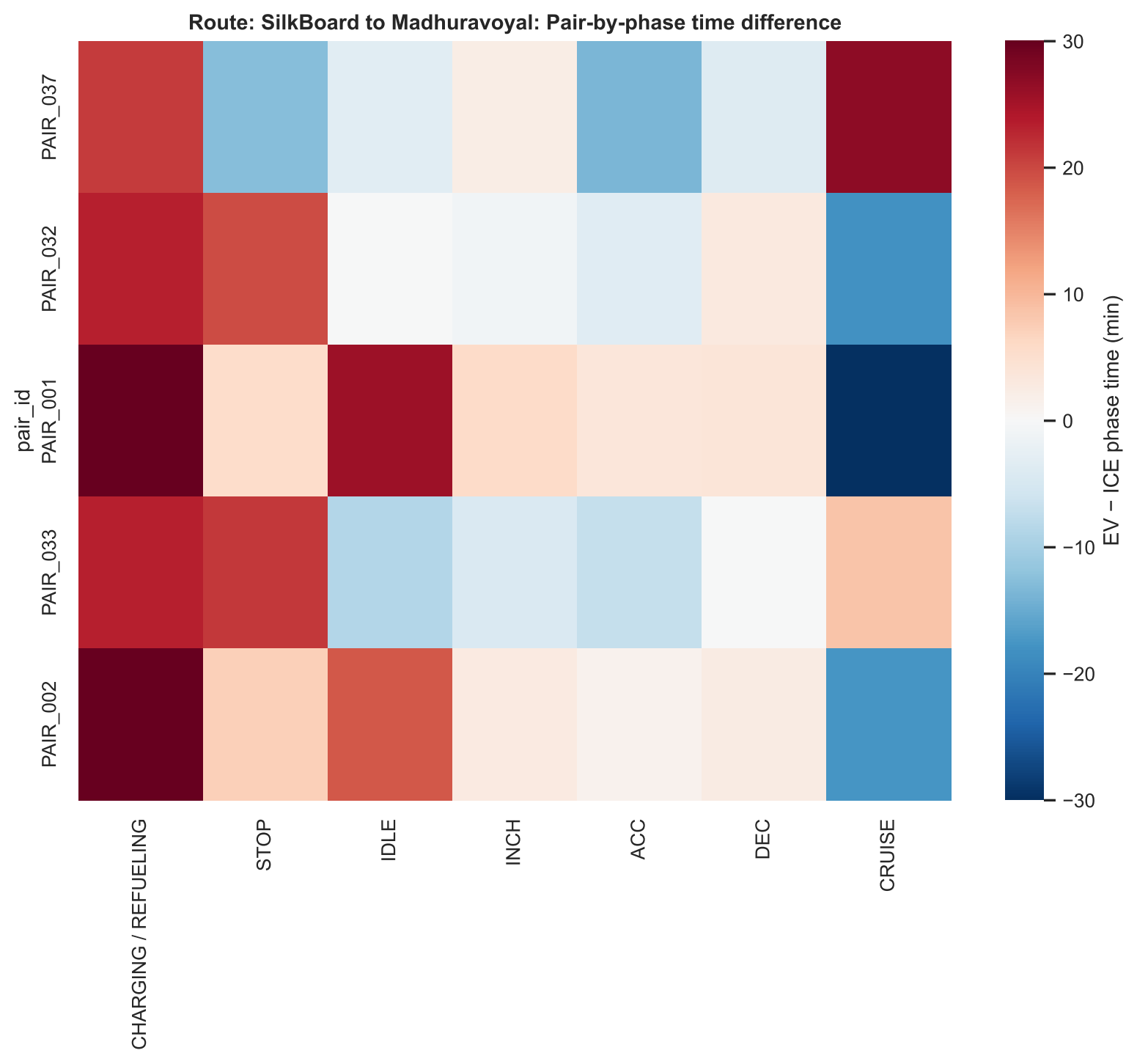
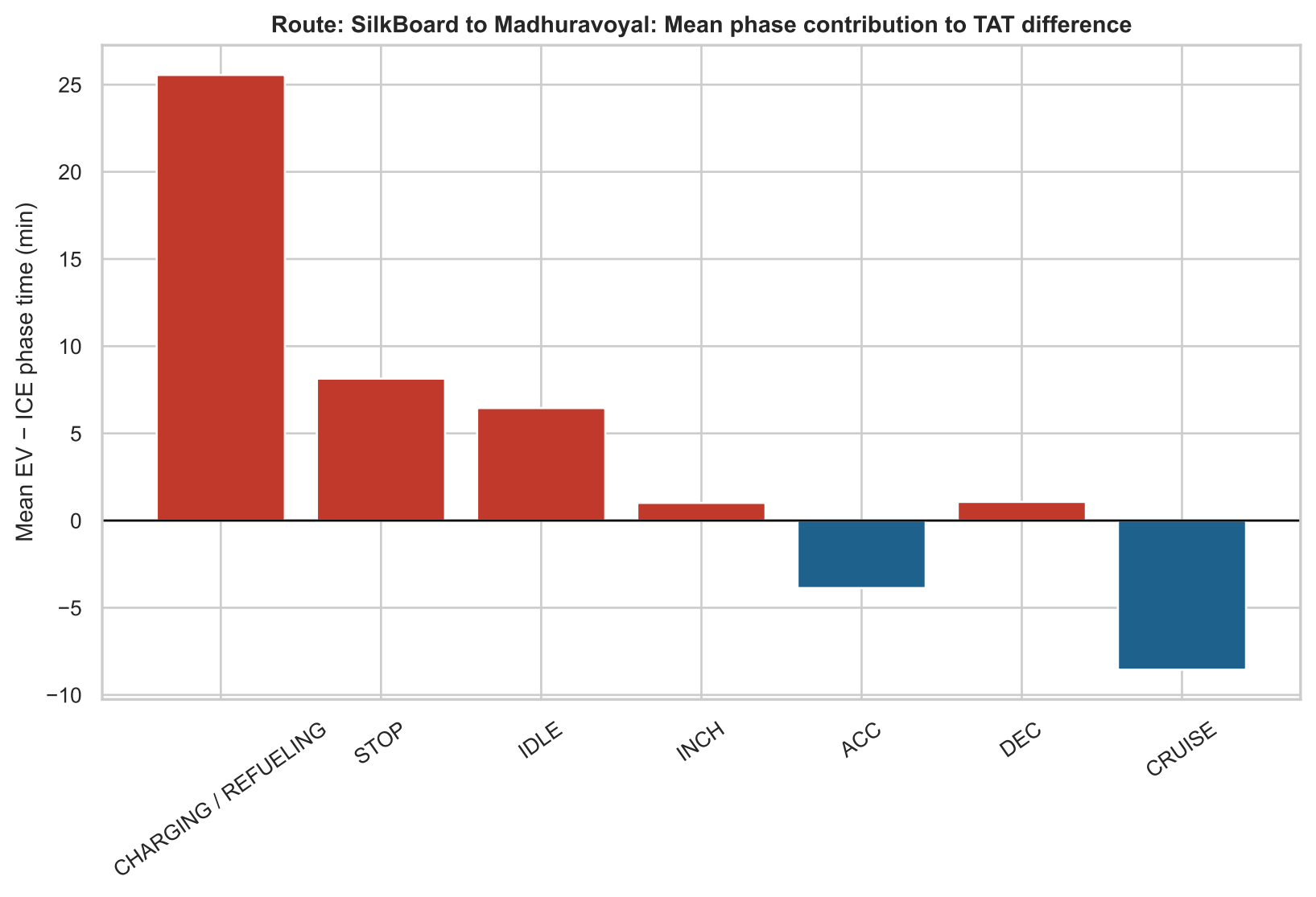
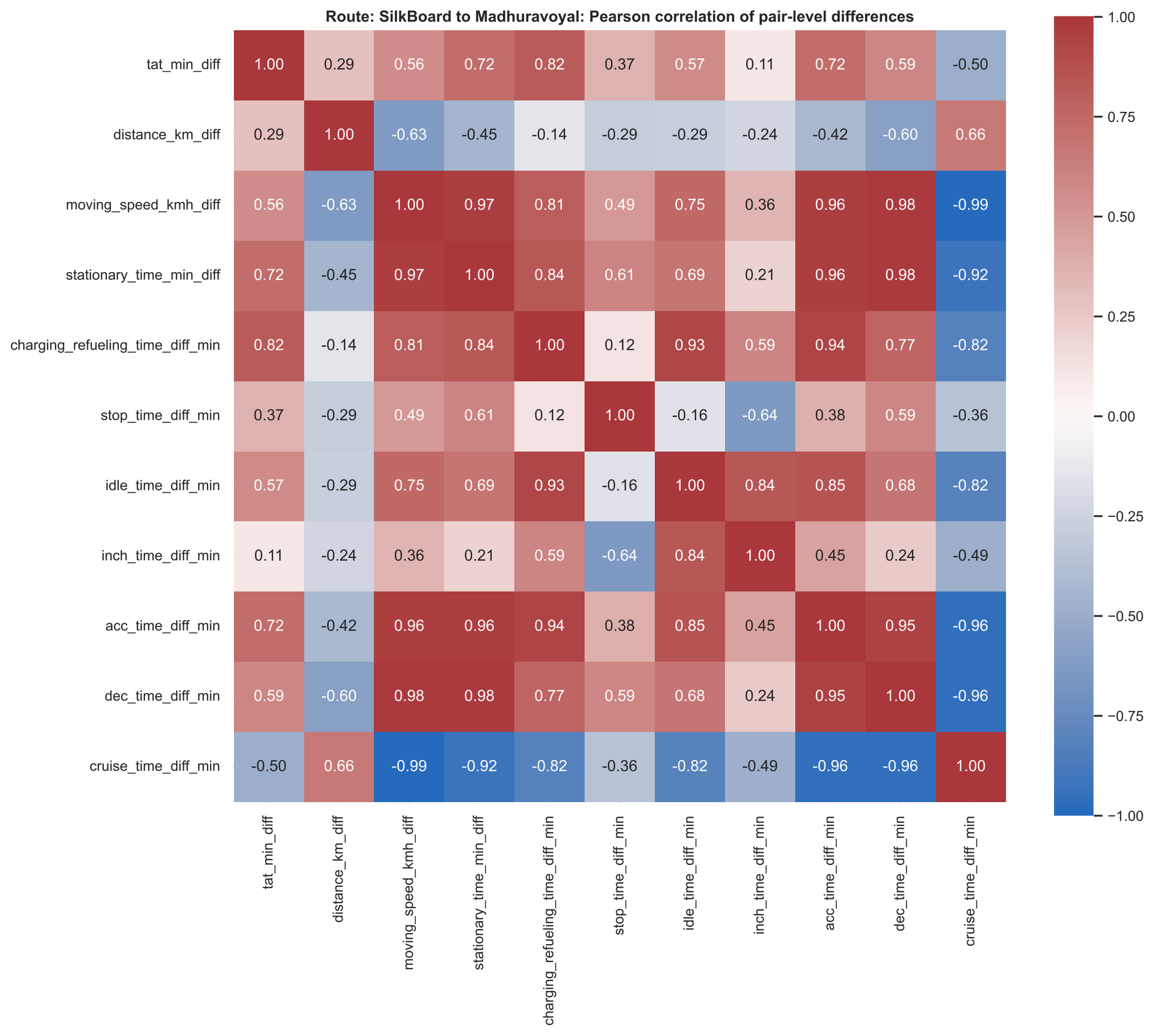

Route: SilkBoard to Madhuravoyal: Ranked pair-level TAT difference
PAIR_002
PAIR_033
PAIR_001
PAIR_032
PAIR_037
0
10
20
30
40
EV TAT − ICE TAT (min)

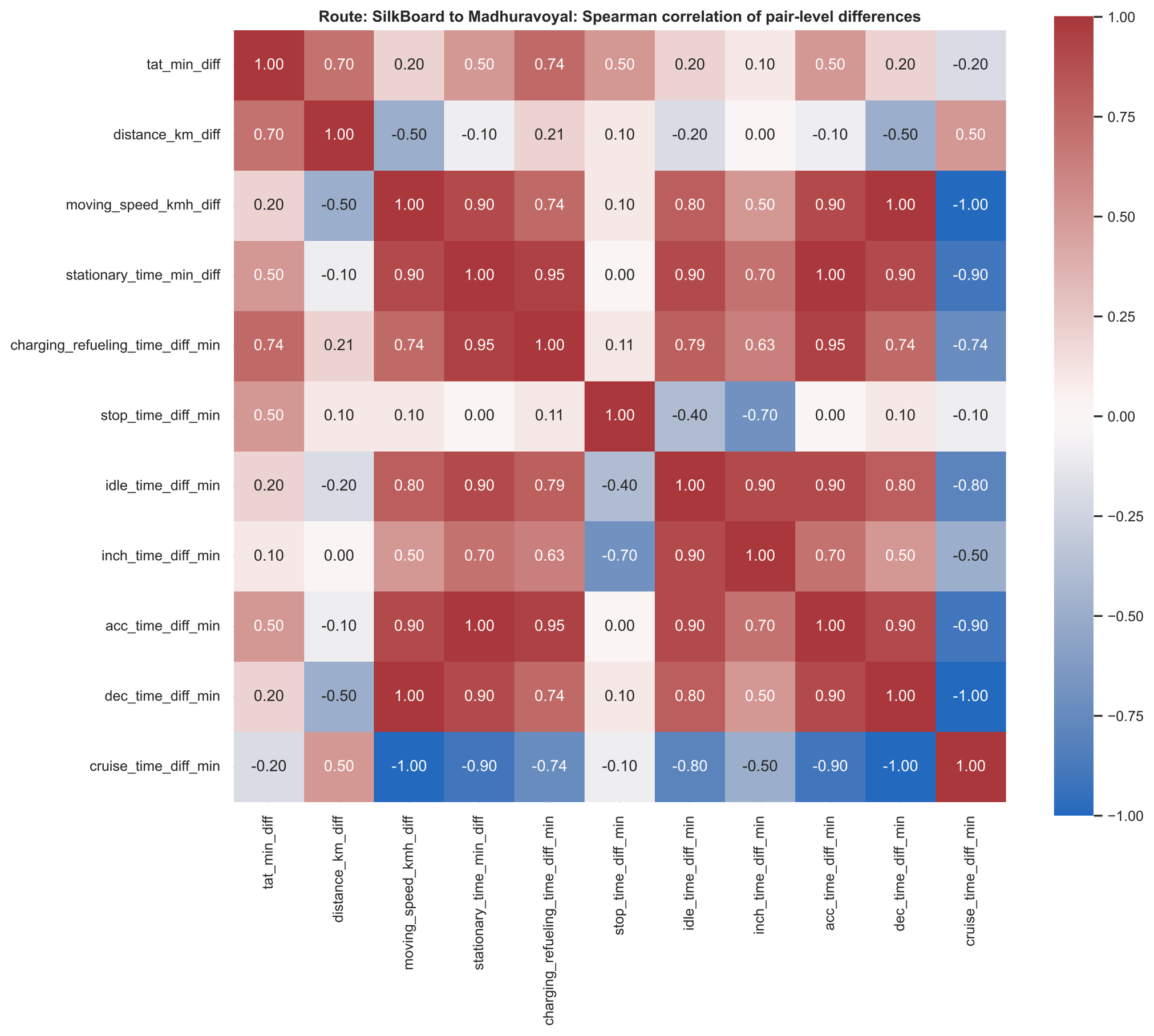

Route: SilkBoard to Madhuravoyal: Distribution of TAT differences
Count
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
10
20
30
40
EV TAT − ICE TAT (min)

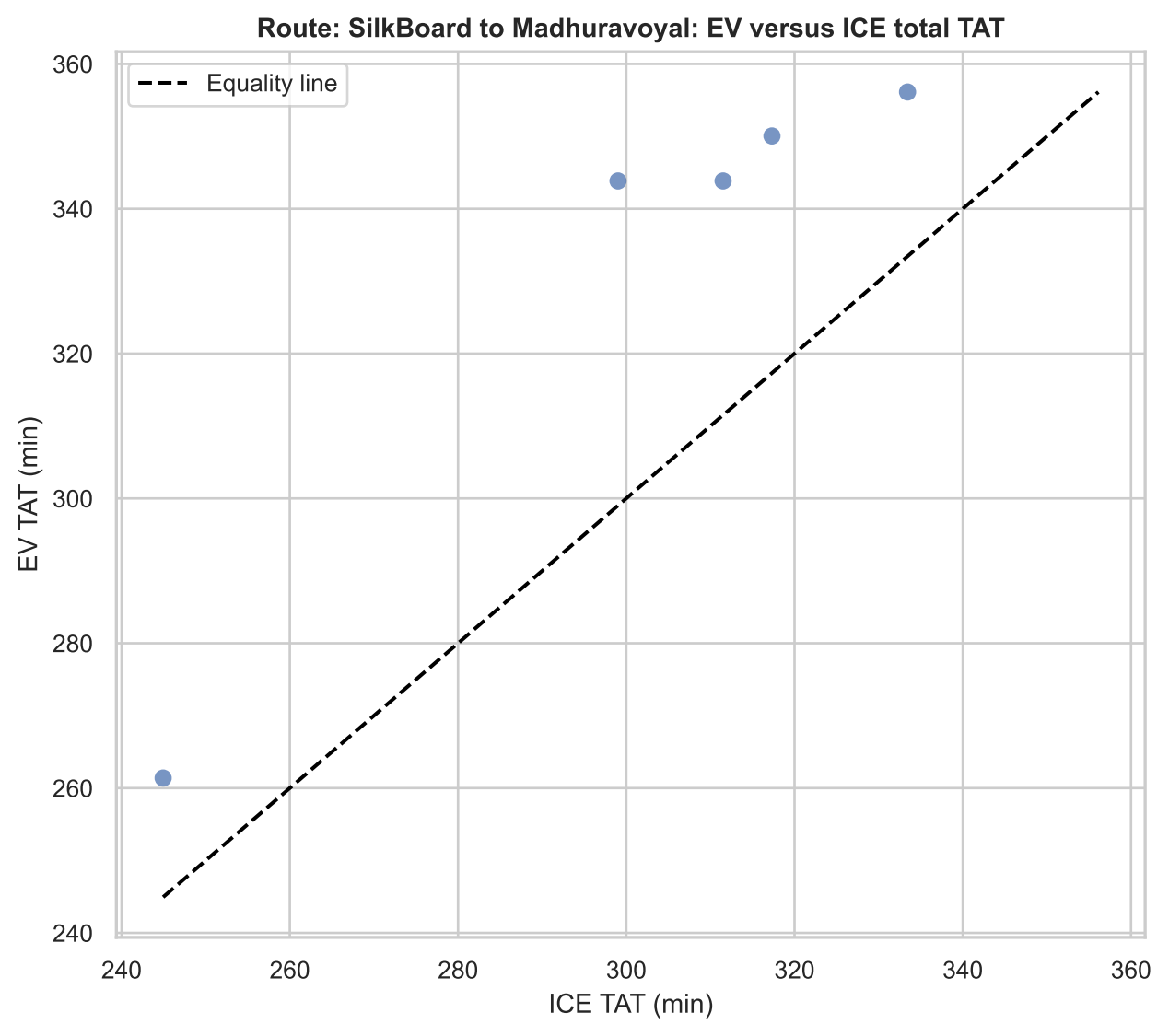

Route: SilkBoard to Madhuravoyal: EV and ICE TAT distribution
Total TAT (min)
360
340
320
300
280
260
240
EV
ICE

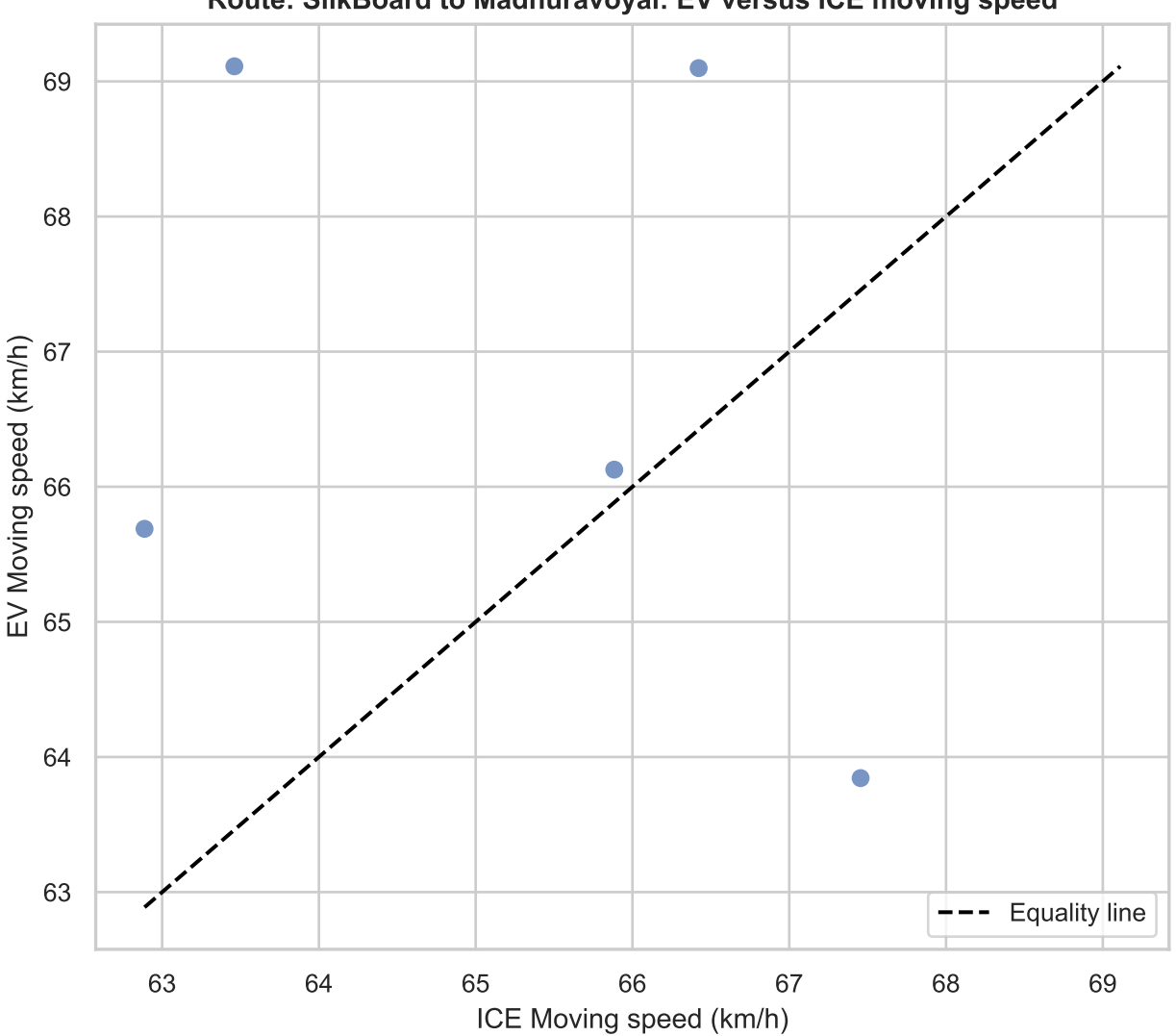

Route: SilkBoard to Madhuravoyal: Pair-by-phase time difference
pair_id
PAIR_037
PAIR_032
PAIR_001
PAIR_033
PAIR_002
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
30
20
10
0
−10
−20
−30

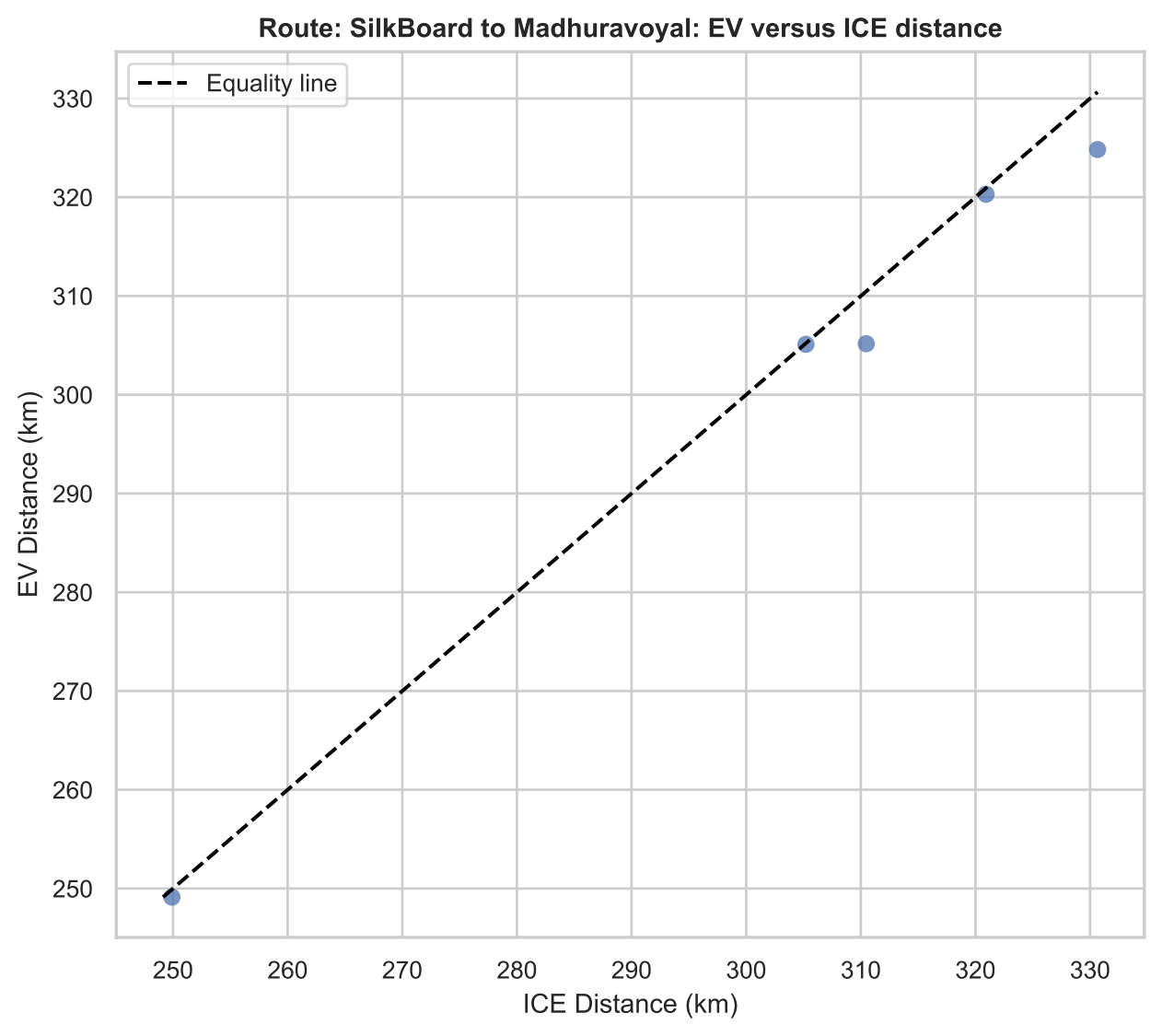

Route: SilkBoard to Madhuravoyal: Mean phase contribution to TAT difference
Mean EV − ICE phase time (min)
25
20
15
10
5
0
−5
−10
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

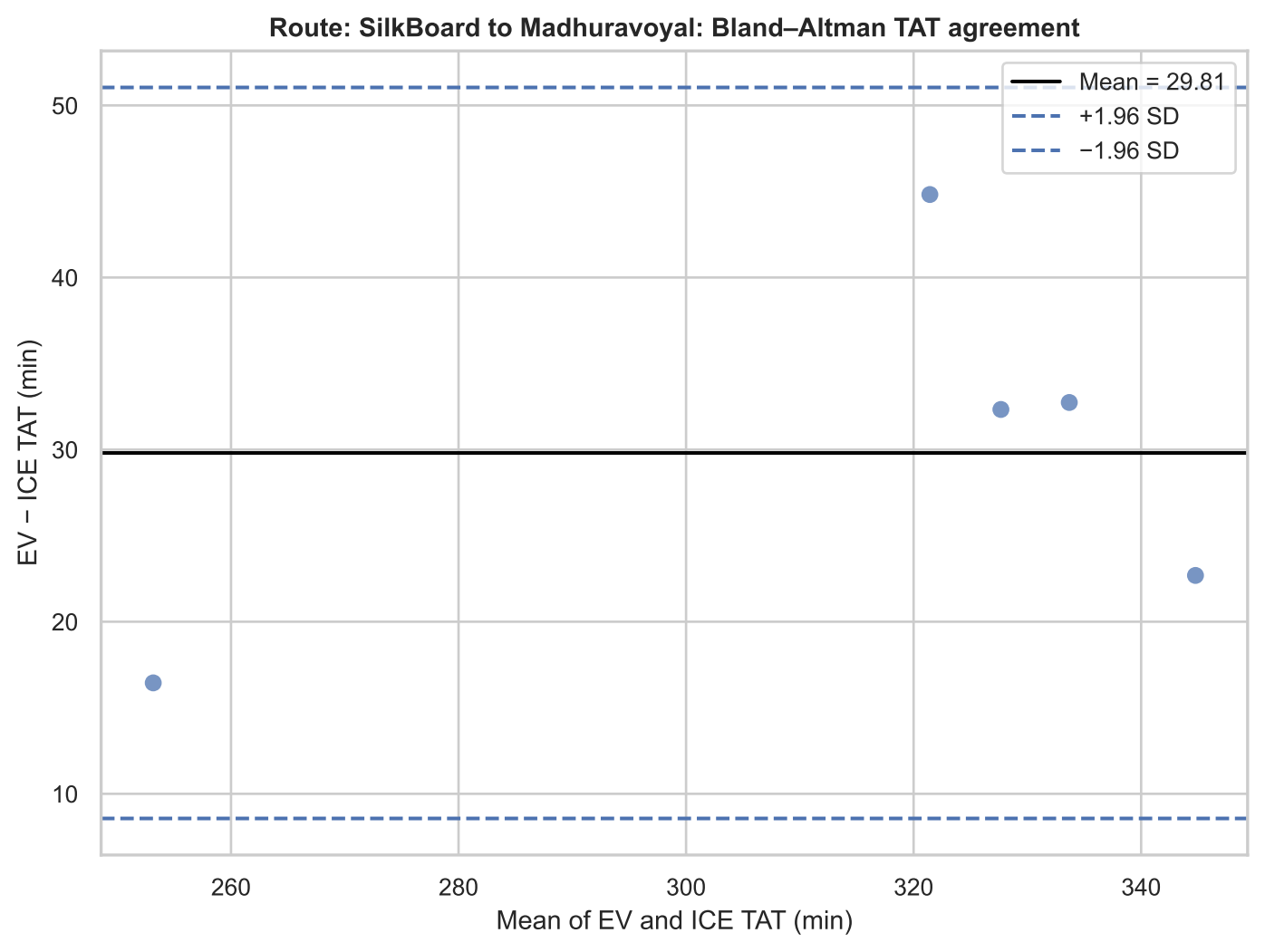

Route: SilkBoard to Madhuravoyal: Pearson correlation of pair-level differences
tat_min_diff 1.00 0.29 0.56 0.72 0.82 0.37 0.57 0.11 0.72 0.59 -0.50
distance_km_diff 0.29 1.00 -0.63 -0.45 -0.14 -0.29 -0.29 -0.24 -0.42 -0.60 0.66
moving_speed_kmh_diff 0.56 -0.63 1.00 0.97 0.81 0.49 0.75 0.36 0.96 0.98 -0.99
stationary_time_min_diff 0.72 -0.45 0.97 1.00 0.84 0.61 0.69 0.21 0.96 0.98 -0.92
charging_refueling_time_diff_min 0.82 -0.14 0.81 0.84 1.00 0.12 0.93 0.59 0.94 0.77 -0.82
stop_time_diff_min 0.37 -0.29 0.49 0.61 0.12 1.00 -0.16 -0.64 0.38 0.59 -0.36
idle_time_diff_min 0.57 -0.29 0.75 0.69 0.93 -0.16 1.00 0.84 0.85 0.68 -0.82
inch_time_diff_min 0.11 -0.24 0.36 0.21 0.59 -0.64 0.84 1.00 0.45 0.24 -0.49
acc_time_diff_min 0.72 -0.42 0.96 0.96 0.94 0.38 0.85 0.45 1.00 0.95 -0.96
dec_time_diff_min 0.59 -0.60 0.98 0.98 0.77 0.59 0.68 0.24 0.95 1.00 -0.96
cruise_time_diff_min -0.50 0.66 -0.99 -0.92 -0.82 -0.36 -0.82 -0.49 -0.96 -0.96 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

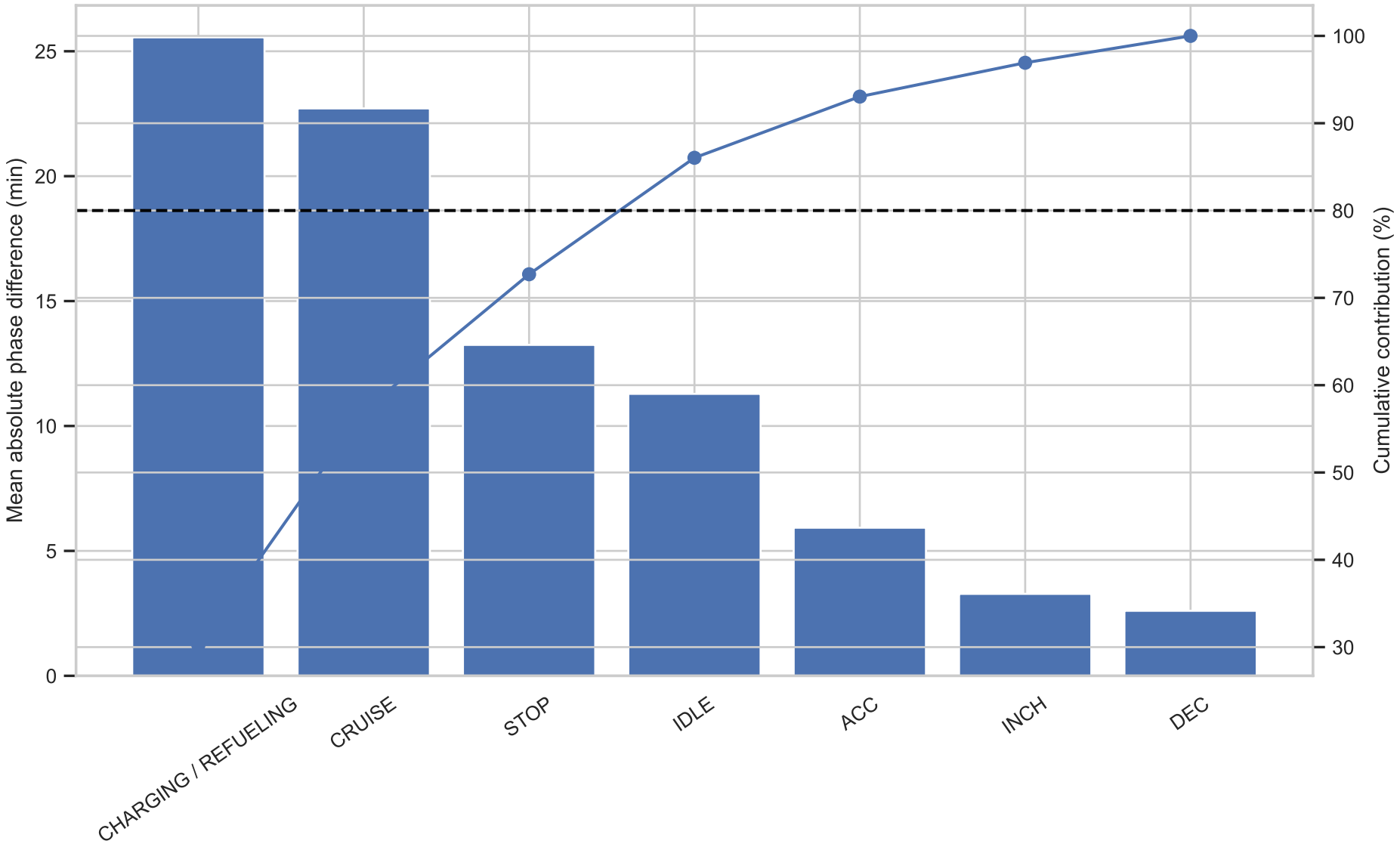

Route: SilkBoard to Madhuravoyal: Spearman correlation of pair-level differences
tat_min_diff 1.00 0.70 0.20 0.50 0.74 0.50 0.20 0.10 0.50 0.20 -0.20
distance_km_diff 0.70 1.00 -0.50 -0.10 0.21 0.10 -0.20 0.00 -0.10 -0.50 0.50
moving_speed_kmh_diff 0.20 -0.50 1.00 0.90 0.74 0.10 0.80 0.50 0.90 1.00 -1.00
stationary_time_min_diff 0.50 -0.10 0.90 1.00 0.95 0.00 0.90 0.70 1.00 0.90 -0.90
charging_refueling_time_diff_min 0.74 0.21 0.74 0.95 1.00 0.11 0.79 0.63 0.95 0.74 -0.74
stop_time_diff_min 0.50 0.10 0.10 0.00 0.11 1.00 -0.40 -0.70 0.00 0.10 -0.10
idle_time_diff_min 0.20 -0.20 0.80 0.90 0.79 -0.40 1.00 0.90 0.90 0.80 -0.80
inch_time_diff_min 0.10 0.00 0.50 0.70 0.63 -0.70 0.90 1.00 0.70 0.50 -0.50
acc_time_diff_min 0.50 -0.10 0.90 1.00 0.95 0.00 0.90 0.70 1.00 0.90 -0.90
dec_time_diff_min 0.20 -0.50 1.00 0.90 0.74 0.10 0.80 0.50 0.90 1.00 -1.00
cruise_time_diff_min -0.20 0.50 -1.00 -0.90 -0.74 -0.10 -0.80 -0.50 -0.90 -1.00 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

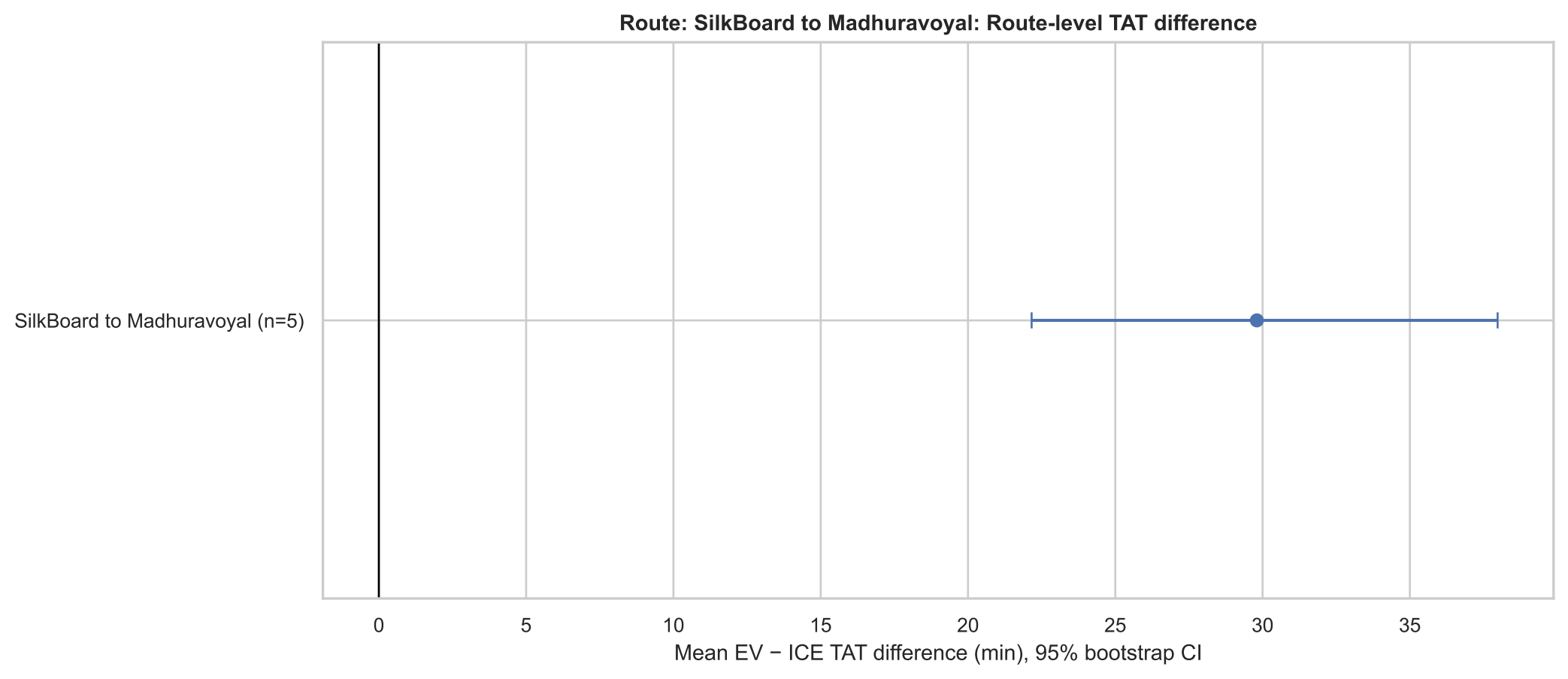

Route: SilkBoard to Madhuravoyal: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
240
260
280
300
320
340
360

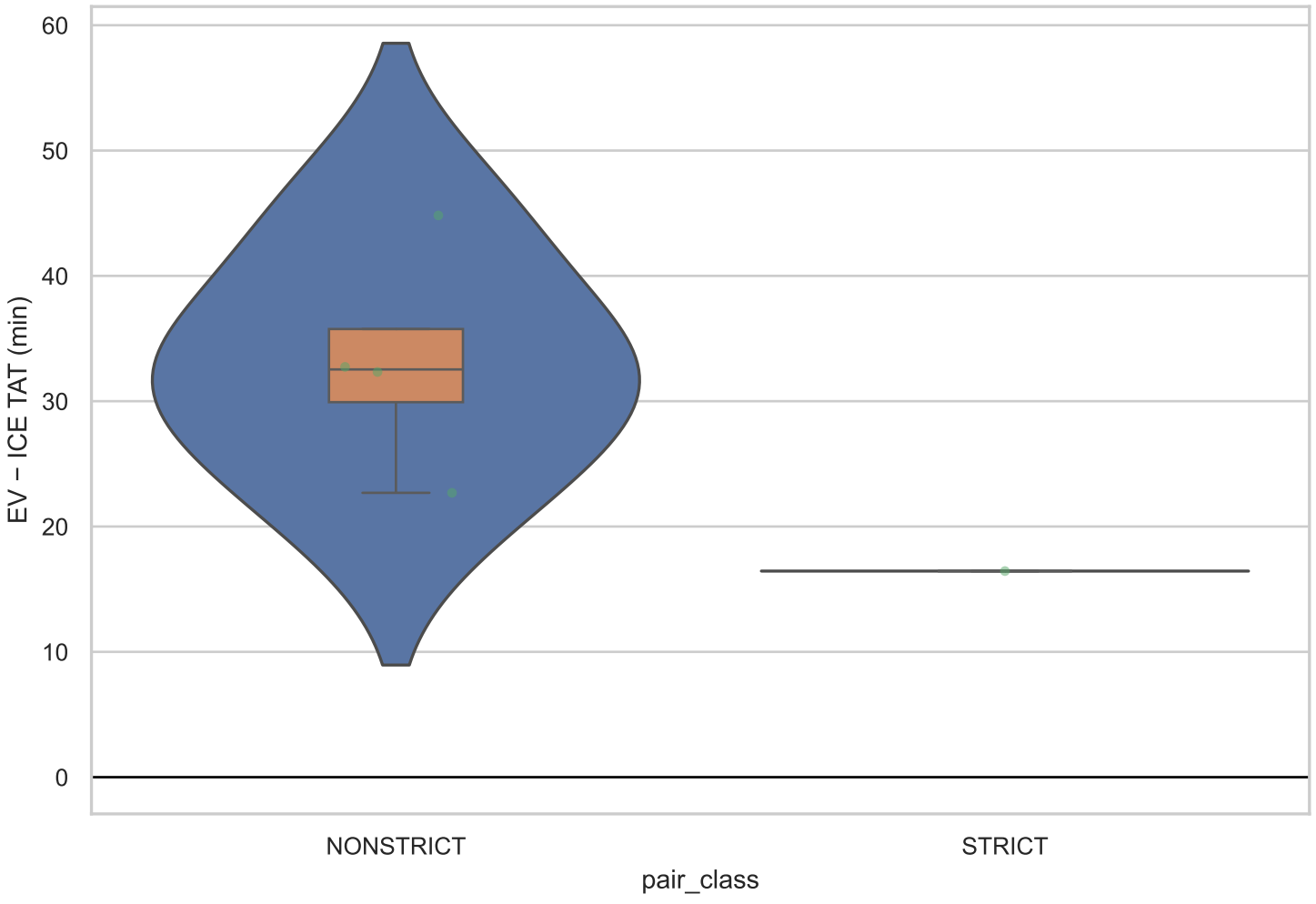

Route: SilkBoard to Madhuravoyal: EV versus ICE moving speed
EV Moving speed (km/h)
ICE Moving speed (km/h)
63
64
65
66
67
68
69
Equality line

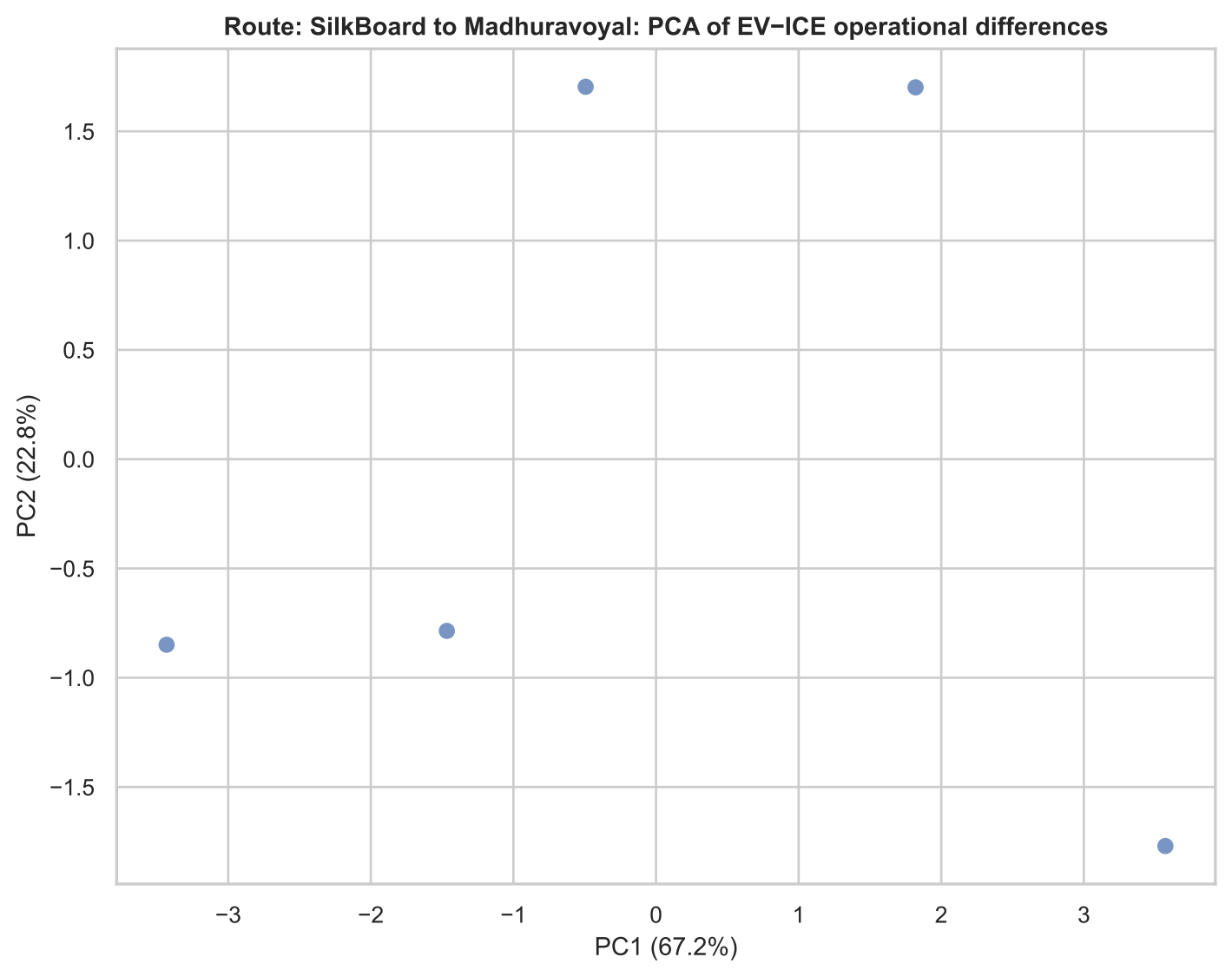

Route: SilkBoard to Madhuravoyal: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
250
260
270
280
290
300
310
320
330

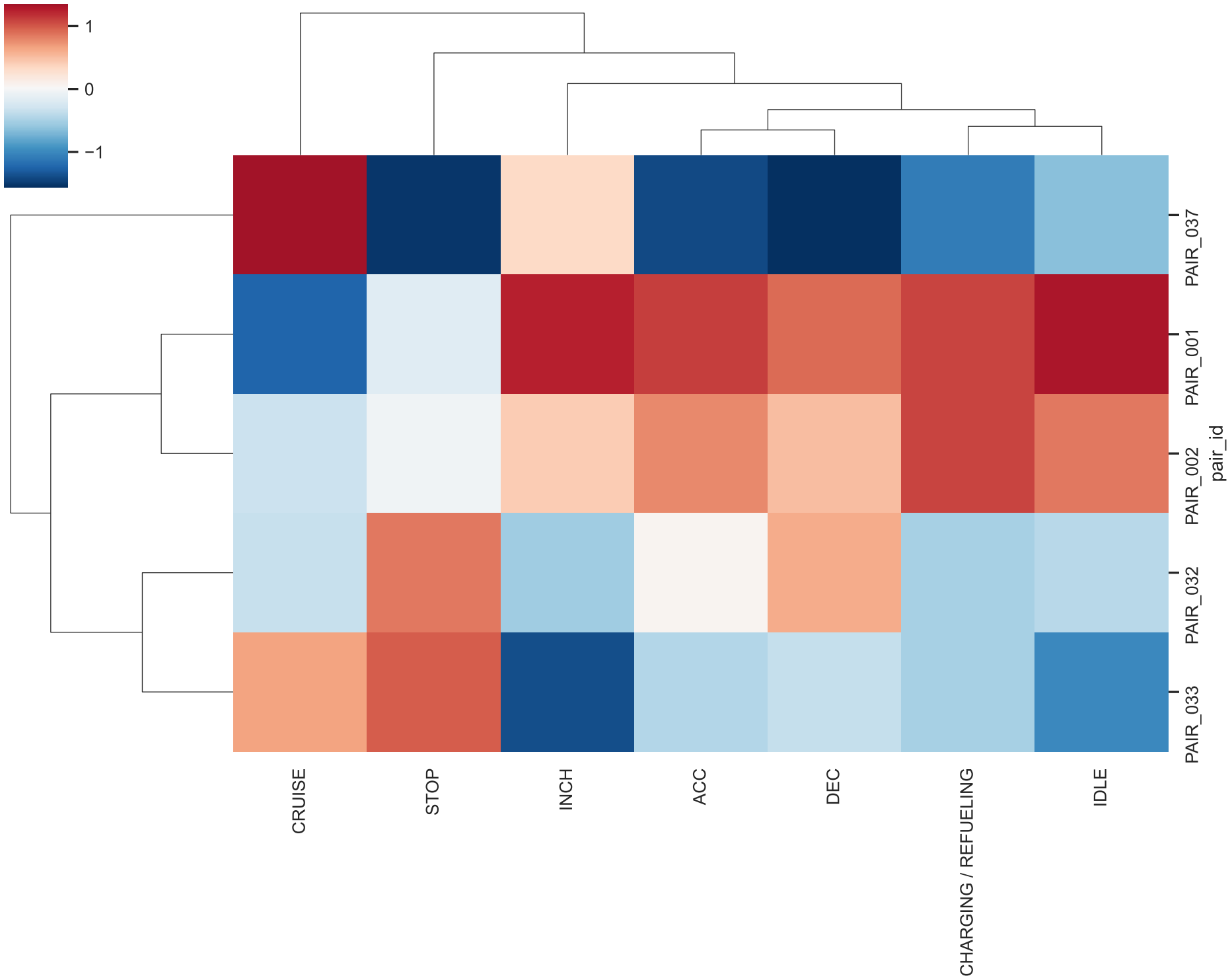

Route: SilkBoard to Madhuravoyal: Bland–Altman TAT agreement
Mean = 29.81
+1.96 SD
−1.96 SD
50
40
30
20
10
EV – ICE TAT (min)
260
280
300
320
340
Mean of EV and ICE TAT (min)


Route: SilkBoard to Madhuravoyal: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
5
10
15
20
25
30
40
50
60
70
80
90
100
CHARGING / REFUELING
CRUISE
STOP
IDLE
ACC
INCH
DEC


Route: SilkBoard to Madhuravoyal: Route-level TAT difference
SilkBoard to Madhuravoyal (n=5)
0
5
10
15
20
25
30
35
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: SilkBoard to Madhuravoyal: STRICT vs NONSTRICT TAT difference
60
50
40
30
20
10
0
EV – ICE TAT (min)
NONSTRICT
STRICT
pair_class


Route: SilkBoard to Madhuravoyal: PCA of EV−ICE operational differences
PC2 (22.8%)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−3
−2
−1
0
1
2
3
PC1 (67.2%)


Route: SilkBoard to Madhuravoyal: Hierarchical clustering of phase-difference signatures
1
0
−1
PAIR_037
PAIR_001
PAIR_002
PAIR_032
PAIR_033
pair_id
CRUISE
STOP
INCH
ACC
DEC
CHARGING / REFUELING
IDLE